AWS – NAT GW, EIG, RT

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Creating Multiple Route Tables.

- We could have multiple Route Tables for Different Subnets in an Single VPC.
- A subnet can be part of a single Route TABLE only.

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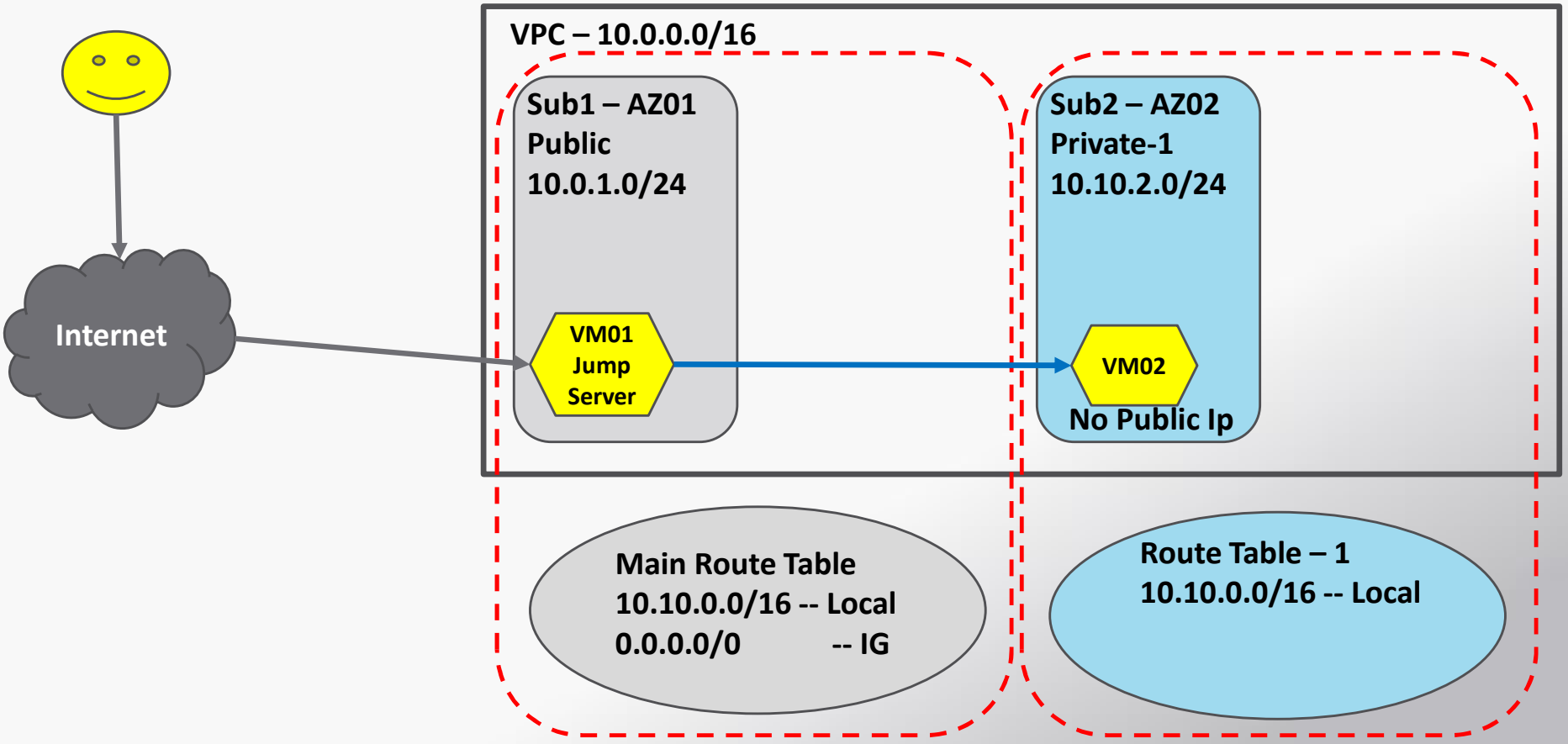
Multiple Route Table Scenario



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IP Address details of EC2 (VM)

- **Private IPv4** → The assigned IPV4 to the VM is Static IP, and does not change even if the VM is shutdown or restarted and stays until the VM is "**Terminated**"
- **Public IPv4** → This is a dynamic IP **allocated** to the VM. It would change once the Vm shutdown and start back. But the IP would remain if the VM is restarted.
  - The public IPv4 is not assigned to the VM, the VPC maintains the "**NAT**" rule for private to Public mapping,
- **" Public IPv6** → This is an static IP, assigned to the VM and stays until the VM is **"terminated**

- **Static Public IPv4 –** Create an **Elastic IP and assign(Associate) the VM to public ip.**
  - This is CHARGED per Hour.

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Different Gateways in VPC

- **Internet Gateway →**
  - It provides an path to the internet for the Virtual Machines in the VPC.
  - The Traffic is 2 ways, both inbound and outbound is allowed for IPv4 and IPv6.
- **Egress only Internet Gateway →**
  - It provides an path to the internet for the Virtual Machines in the VPC on IPv6 only.
  - The Traffic is **One way**, only outbound is allowed for IPv6 from the Virtual Machine.
- **NAT Gateway →**
  - It provides an path to the internet for the Virtual Machines in the VPC on **IPv4 only**.
  - The Traffic is **One way**, **only outbound is allowed** for IPv4 from the Virtual Machine.
  - There is an **Public IP assigned on the Gateway**, and there is NO USE of public ip on the Virtual machine.
  - This is NOT available for IPv6, as “NAT Gateway” is used to conserve IPv4.
  - We cannot PING any of the public ip from the VM.

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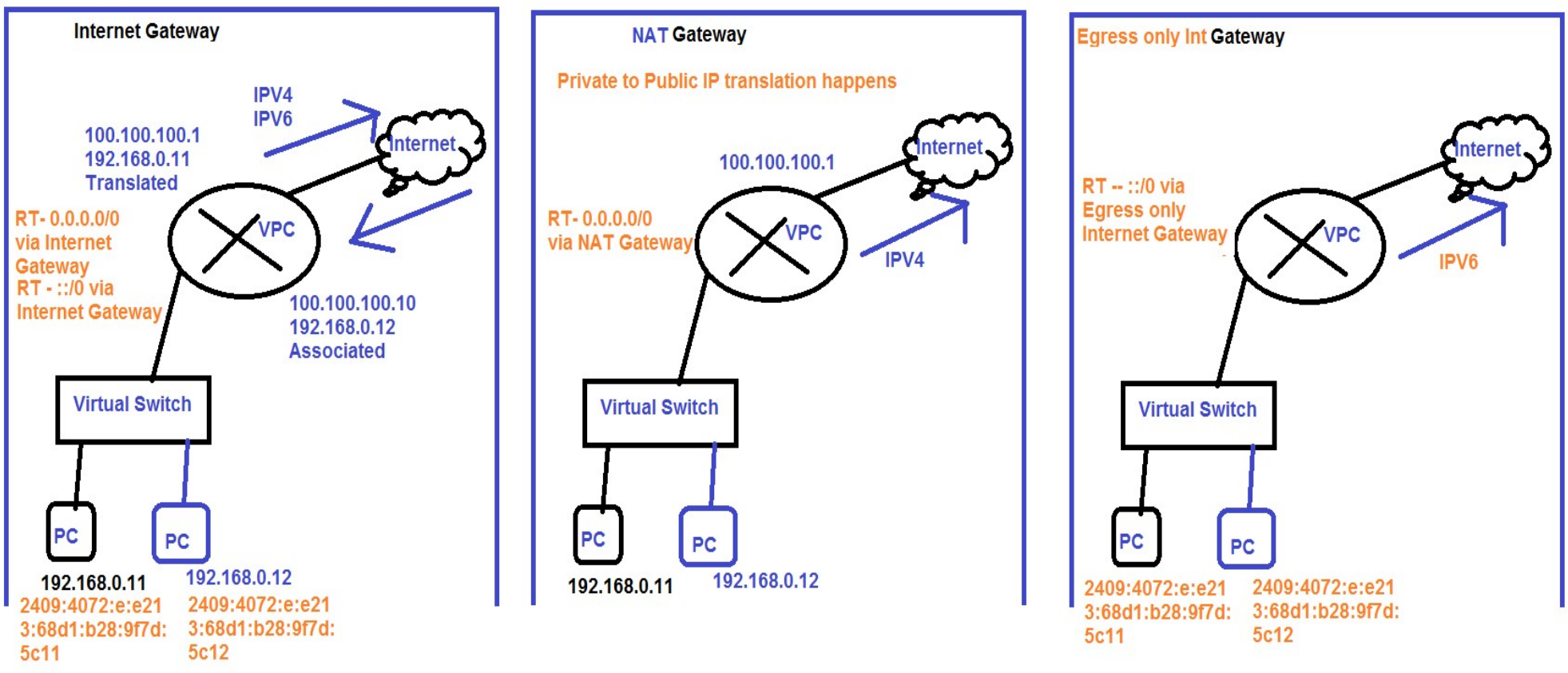
Internet GW vs NAT GW vs Egress GW



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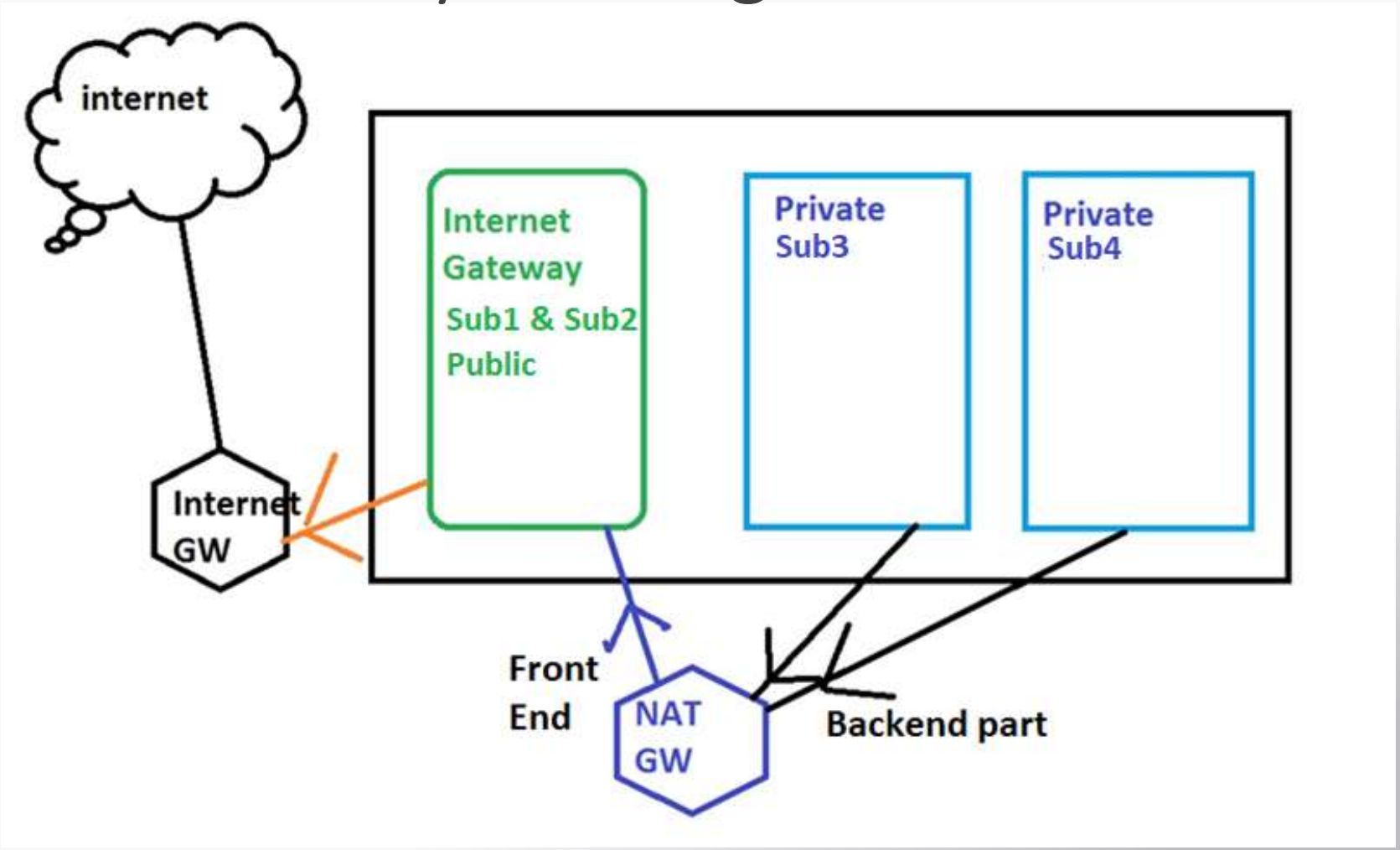
NAT Gateway Working



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Steps to Create NAT Gateway

1. **Create an separate routing table for the private Subnet and associate the subnet.**
2. **Create an NAT Gateway assigning it to the Public subnet for the Internet access and assign an Elastic IP to the same**
3. **Update the custom Routing table created in Step1, with default route pointing towards NAT Gateway.**

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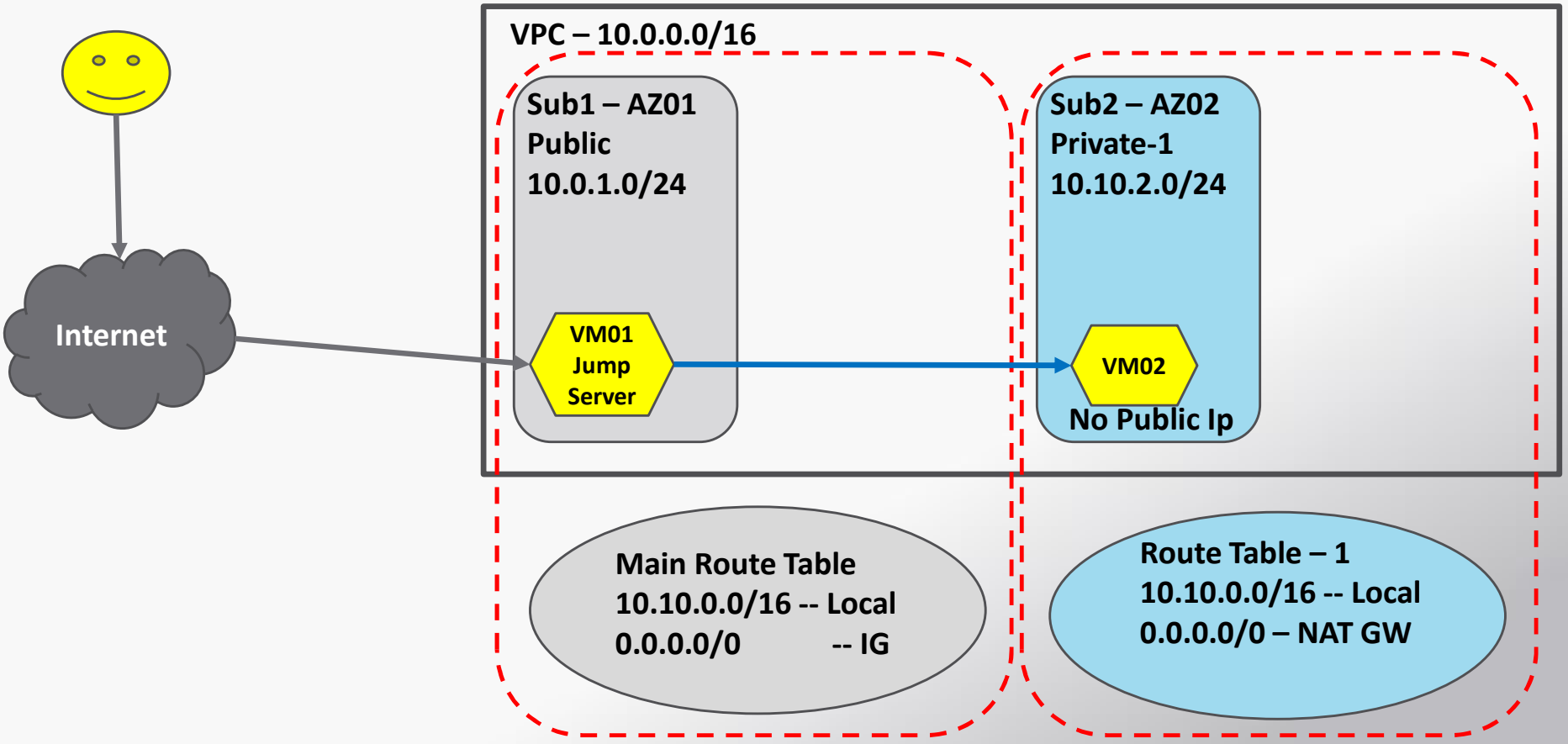
Multiple Route Table Scenario



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PRICING

- **VPC – NOT Charged**
- **Subnets – NOT Charged**
- **Internet Gateway – NOT Charged**
- **Routing Table – NOT Charged**
- **NAT Gateway – Charged, as there is an Public IPv4 assigned to it.**
- **Egress only IG – NOT Charged**
- **VPN -- Charged, as there is an Public IPv4 assigned to it.**

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Elastic IP pricing

An Elastic IP address doesn’t incur charges as long as the following conditions are true:

- The Elastic IP address is associated with an EC2 instance.
- The instance associated with the Elastic IP address is running.
- The instance has only one Elastic IP address attached to it.

You're charged by the hour for each Elastic IP address that doesn't meet these conditions.

Region: US East (N. Virginia)

- $0.005 per additional IP address associated with a running instance per hour on a pro rata basis
- $0.005 per Elastic IP address not associated with a running instance per hour on a pro rata basis

- $0.00 per Elastic IP address remap for the first 100 remaps per month
- $0.10 per Elastic IP address remap for additional remaps over 100 per month

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Troubleshooting VPC

Basic Troubleshooting steps if the EC2 instance is not getting connected.

- Check Weather “Internet gateway” is created an assigned to “Routing Table”.
- If custom Route table created, weather “Subnet’s” are associated to the new Routing table.
- Weather “PORTS” are allowed in the security group for “inbound” and “outbound”.

- https://aws.amazon.com/premiumsupport/knowledge-center/troubleshoot-vpc-route-table/

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